

# MLOps Research Study

## Automated Research Summary Report

This compact report collects the principal configuration, validation, generalization, learning-curve, and permutation-sensitivity outputs produced by the current MLOps run. Detailed CSV/JSON files remain the authoritative source for full results.

| Setting                | Value                                |
|------------------------|--------------------------------------|
| Generated              | 2026-08-23T10:59:22+03:30            |
| Dataset                | G:\Researching\GT_GL_MLOps\GTFS_M... |
| Dataset SHA256         | daf22e9b3887a8070242eec567c90206a... |
| Targets                | AC                                   |
| Grouping enabled       | False                                |
| Group column           | Location_No                          |
| Train / test           | 0.8/0.2                              |
| CV splits / repeats    | 5/3                                  |
| Classification enabled | True                                 |
| Class boundaries       | {"AC": {"lower": 0.75, "upper": 1... |

## Analysis availability

| Analysis                       | Status    | Files |
|--------------------------------|-----------|-------|
| Data quality                   | Available | 5     |
| Multicollinearity              | Available | 4     |
| Regression nested CV           | Available | 48    |
| Regression learning curves     | Available | 46    |
| Regression permutation         | Available | 48    |
| Classification nested CV       | Available | 36    |
| Classification learning curves | Available | 36    |
| Classification permutation     | Available | 36    |

Selected / representative model results

| Task           | Target | Feature set | Model | Metric   | CV    | CV SD | Test  | Gap   |
|----------------|--------|-------------|-------|----------|-------|-------|-------|-------|
| regression     | AC     | fs1         | SVR   | R2       | 0.260 | 0.259 | 0.358 | 0.150 |
| regression     | AC     | fs2         | SVR   | R2       | 0.279 | 0.242 | 0.341 | 0.158 |
| regression     | AC     | fs3         | SVR   | R2       | 0.284 | 0.255 | 0.318 | 0.173 |
| regression     | AC     | fs4         | SVR   | R2       | 0.096 | 0.166 | 0.186 | 0.123 |
| classification | AC     | fs1         | XGB   | Macro-F1 | 0.649 | 0.105 | 0.784 | 0.068 |
| classification | AC     | fs2         | SVC   | Macro-F1 | 0.619 | 0.150 | 0.675 | 0.040 |
| classification | AC     | fs3         | SVC   | Macro-F1 | 0.575 | 0.156 | 0.644 | 0.101 |
| classification | AC     | fs4         | ANN   | Macro-F1 | 0.500 | 0.103 | 0.563 | 0.063 |

# Nested cross-validation

Nested CV is summarized from the saved outer-fold files. Hyperparameter optimization remains inside the inner CV loop in the training programs; the table below reports the outer-fold generalization estimates when available.

| Task           | Target | Feature set | Model  | Outer folds | Metric   | Mean  | SD    | Secondary         | Value |
|----------------|--------|-------------|--------|-------------|----------|-------|-------|-------------------|-------|
| classification | AC     | fs1         | ANN    | 5           | Macro-F1 | 0.622 | 0.113 | Balanced accuracy | 0.644 |
| classification | AC     | fs2         | ANN    | 5           | Macro-F1 | 0.549 | 0.054 | Balanced accuracy | 0.553 |
| classification | AC     | fs3         | LOGREG | 5           | Macro-F1 | 0.524 | 0.056 | Balanced accuracy | 0.559 |
| classification | AC     | fs4         | ANN    | 5           | Macro-F1 | 0.479 | 0.166 | Balanced accuracy | 0.525 |
| regression     | AC     | fs2         | LASSO  | 5           | R2       | 0.259 | 0.387 | RMSE              | 0.333 |
| regression     | AC     | fs1         | ANN    | 5           | R2       | 0.188 | 0.314 | RMSE              | 0.352 |
| regression     | AC     | fs3         | ET     | 5           | R2       | 0.103 | 0.168 | RMSE              | 0.383 |
| regression     | AC     | fs4         | GBR    | 5           | R2       | 0.034 | 0.135 | RMSE              | 0.402 |

# Learning-curve analysis

| Task       | Target | Feature set | Model | Final CV R2 | Final CV F1 | Final gap | Trend     | More-data signal |
|------------|--------|-------------|-------|-------------|-------------|-----------|-----------|------------------|
| regression | AC     | fs1         | SVR   | 0.260       | -           | 0.142     | improving | True             |
| regression | AC     | fs1         | GBR   | 0.228       | -           | 0.215     | improving | False            |
| regression | AC     | fs1         | XGB   | 0.169       | -           | 0.297     | improving | False            |
| regression | AC     | fs1         | DT    | 0.147       | -           | 0.262     | improving | True             |
| regression | AC     | fs1         | ET    | 0.142       | -           | 0.270     | improving | True             |
| regression | AC     | fs1         | EN    | 0.134       | -           | 0.233     | improving | False            |
| regression | AC     | fs1         | ANN   | 0.129       | -           | 0.247     | improving | True             |
| regression | AC     | fs1         | RIDGE | 0.117       | -           | 0.217     | improving | True             |
| regression | AC     | fs1         | LASSO | 0.108       | -           | 0.277     | improving | False            |
| regression | AC     | fs1         | MLR   | 0.097       | -           | 0.289     | improving | False            |
| regression | AC     | fs1         | RF    | 0.069       | -           | 0.427     | improving | False            |
| regression | AC     | fs2         | SVR   | 0.279       | -           | 0.151     | improving | True             |
| regression | AC     | fs2         | ANN   | 0.261       | -           | 0.197     | improving | False            |

| Task       | Target | Feature set | Model | Final CV R2 | Final CV F1 | Final gap | Trend     | More-data signal |
|------------|--------|-------------|-------|-------------|-------------|-----------|-----------|------------------|
| regression | AC     | fs2         | EN    | 0.260       | -           | 0.176     | improving | False            |
| regression | AC     | fs2         | LASSO | 0.260       | -           | 0.176     | improving | False            |
| regression | AC     | fs2         | RIDGE | 0.256       | -           | 0.180     | improving | False            |
| regression | AC     | fs2         | MLR   | 0.248       | -           | 0.189     | improving | False            |
| regression | AC     | fs2         | GBR   | 0.232       | -           | 0.244     | improving | True             |
| regression | AC     | fs2         | ET    | 0.183       | -           | 0.277     | improving | True             |
| regression | AC     | fs2         | XGB   | 0.164       | -           | 0.399     | improving | True             |
| regression | AC     | fs2         | RF    | 0.133       | -           | 0.416     | improving | False            |
| regression | AC     | fs2         | DT    | 0.032       | -           | 0.479     | improving | False            |
| regression | AC     | fs2         | KNN   | -0.032      | -           | 0.674     | improving | True             |
| regression | AC     | fs3         | SVR   | 0.284       | -           | 0.173     | improving | True             |

# Permutation sensitivity / feature importance

Permutation sensitivity is computed by perturbing one feature at a time and measuring the decrease in the configured predictive score. Large positive magnitudes indicate greater dependence of the fitted model on that predictor; negative values can occur and should not be interpreted as beneficial causal effects.

| Task       | Target | Feature set | Model | Rank | Feature   | Mean importance | Relative % |
|------------|--------|-------------|-------|------|-----------|-----------------|------------|
| regression | AC     | fs1         | ANN   | 1    | PI/FF     | 0.610           | 99.226     |
| regression | AC     | fs1         | ANN   | 2    | Ucs_class | -0.005          | 0.774      |
| regression | AC     | fs1         | DT    | 1    | PI/FF     | 0.879           | 100.000    |
| regression | AC     | fs1         | DT    | 2    | Ucs_class | 0.000           | 0.000      |
| regression | AC     | fs1         | EN    | 1    | PI/FF     | 0.649           | 100.000    |
| regression | AC     | fs1         | EN    | 2    | Ucs_class | 0.000           | 0.000      |
| regression | AC     | fs1         | ET    | 1    | PI/FF     | 0.636           | 99.364     |
| regression | AC     | fs1         | ET    | 2    | Ucs_class | 0.004           | 0.636      |
| regression | AC     | fs1         | GBR   | 1    | PI/FF     | 0.862           | 100.000    |
| regression | AC     | fs1         | GBR   | 2    | Ucs_class | 0.000           | 0.000      |
| regression | AC     | fs1         | KNN   | 1    | PI/FF     | 0.940           | 96.831     |
| regression | AC     | fs1         | KNN   | 2    | Ucs_class | 0.031           | 3.169      |
| regression | AC     | fs1         | LASSO | 1    | PI/FF     | 0.857           | 98.046     |
| regression | AC     | fs1         | LASSO | 2    | Ucs_class | 0.017           | 1.954      |
| regression | AC     | fs1         | MLR   | 1    | PI/FF     | 0.959           | 96.560     |
| regression | AC     | fs1         | MLR   | 2    | Ucs_class | 0.034           | 3.440      |
| regression | AC     | fs1         | RF    | 1    | PI/FF     | 0.983           | 97.641     |
| regression | AC     | fs1         | RF    | 2    | Ucs_class | 0.024           | 2.359      |
| regression | AC     | fs1         | RIDGE | 1    | PI/FF     | 0.556           | 99.993     |
| regression | AC     | fs1         | RIDGE | 2    | Ucs_class | -0.000          | 0.007      |
| regression | AC     | fs1         | SVR   | 1    | PI/FF     | 0.733           | 99.748     |
| regression | AC     | fs1         | SVR   | 2    | Ucs_class | 0.002           | 0.252      |
| regression | AC     | fs1         | XGB   | 1    | PI/FF     | 0.818           | 100.000    |
| regression | AC     | fs1         | XGB   | 2    | Ucs_class | 0.000           | 0.000      |

| Task       | Target | Feature set | Model | Rank | Feature   | Mean importance | Relative % |
|------------|--------|-------------|-------|------|-----------|-----------------|------------|
| regression | AC     | fs2         | ANN   | 1    | FF        | 0.508           | 57.240     |
| regression | AC     | fs2         | ANN   | 2    | PI        | 0.337           | 37.989     |
| regression | AC     | fs2         | ANN   | 3    | Ucs_class | 0.042           | 4.771      |
| regression | AC     | fs2         | DT    | 1    | FF        | 0.612           | 66.176     |
| regression | AC     | fs2         | DT    | 2    | PI        | 0.313           | 33.824     |
| regression | AC     | fs2         | DT    | 3    | Ucs_class | 0.000           | 0.000      |
| regression | AC     | fs2         | EN    | 1    | FF        | 0.571           | 69.519     |
| regression | AC     | fs2         | EN    | 2    | PI        | 0.250           | 30.481     |
| regression | AC     | fs2         | EN    | 3    | Ucs_class | 0.000           | 0.000      |
| regression | AC     | fs2         | ET    | 1    | FF        | 0.398           | 61.808     |
| regression | AC     | fs2         | ET    | 2    | PI        | 0.188           | 29.210     |
| regression | AC     | fs2         | ET    | 3    | Ucs_class | 0.058           | 8.982      |

# Reproducibility record

| Item                   | Value                                |
|------------------------|--------------------------------------|
| Python                 | 3.13.2 (tags/v3.13.2:4f8bb39, Feb... |
| Platform               | Windows-10-10.0.19045-SP0            |
| Active params          | G:\Researching\GT_GL_MLops\GTFS_M... |
| Random state           | 42                                   |
| Regression reports     | G:\Researching\GT_GL_MLops\GTFS_M... |
| Classification reports | G:\Researching\GT_GL_MLops\GTFS_M... |
| pkg:numpy              | 2.5.2                                |
| pkg:pandas             | 2.3.3                                |
| pkg:scipy              | 1.18.0                               |
| pkg:scikit-learn       | 1.9.0                                |
| pkg:xgboost            | 3.4.1                                |
| pkg:statsmodels        | 0.14.6                               |

| Item           | Value  |
|----------------|--------|
| pkg:matplotlib | 3.11.1 |
| pkg:joblib     | 1.5.3  |
| pkg:PyYAML     | 6.0.3  |
| pkg:reportlab  | 5.0.1  |

## Interpretation note

The independent test partition should be interpreted as final holdout evidence and not as a tuning criterion. Nested CV, repeated/group-aware CV, learning curves, and permutation sensitivity answer different questions and should be considered jointly rather than collapsed into a single performance number.